

ITBP480: Senior Graduation Project I

Final Project Report

Spring 2026

A Machine Learning-Based System for Predicting Academic Stress Levels in University Students

Project ID: 23

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Executive Summary

University students often face academic pressure due to heavy course loads, tight schedules, and overlapping exams. This pressure It can really impact how they feel, how well they can focus in how they do in university. Many students are not fully aware of how their workload contributes to stress and often lack simple tools to help them manage it effectively. The system focuses on key factors such as credit hours, weekly schedules, study time, and exam load. These are combined with responses from a structured questionnaire to provide a clearer understanding of the student's condition. The proposed solution is a data driven system that collects user input and processes it using a machine learning model. Based on the provided data, the system classifies stress levels into categories such as normal, mild, moderate, or high risk. This allows students to gain insight into their stress levels in a simple and understandable way. The system will consider be efficient and user-friendly. It is Simple Developed using Python for handling data and creating models, and MySQL for database management. for secure data storage, and scikit-learn for building the classification model. These technologies ensure reliability, accuracy, and ease of use. A key feature of the system is its ability to analyze academic workload intelligently. It Evaluates things like how packed a schedule is and when exams are spread out to figure out potential overload. For example,students with high credit hours and multiple exams within a short period are flagged as experiencing higher pressure. This makes the predictions more accurate and gives us more useful information. It seems like the system not only sorts things into categories, but also gives some simple advice on what to do next. These may include improving time management, reducing study overload, and adjusting study habits during exam periods. The system is also designed to be scalable,allowing future improvements such as advanced analytics or integration with web and mobile platforms. The impact of this project lies in raising awareness about the connection between academic workload and mental health. By providing early indicators of stress, it encourages students to take proactive steps and make better academic decisions. It is important to note that the system does not replace professional medical advice but serves as a support It's a valuable tool that helps us in many ways. The people in charge also make sure to keep our personal information safe and private.

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Chapter 1

Introduction

Nowadays, there are plenty of opportunities that are offered thanks to machine learning approaches and big data analytics. Analyzing patterns in student datasets can provide us with important stress-related information and indicators. This research project seeks to design a machine learning system that will estimate the level of students' stress based on academic load factors like credits hours, schedule, and exam distribution among other characteristics. By correlating those parameters with answers provided by a certain questionnaire, one can get a more accurate picture of student's conditions.

The key objective of this tool is to assist students in managing their academic workloads. Time management, studying in an organized manner, and making good decisions on academics are made possible by categorizing the degree of stress faced by the user.

1.1 Problem Statement

University students experience high levels of stress due to academic workload, deadlines, and study pressure. These factors can negatively affect their men tal health, including increased anxiety and reduced focus. Therefore, there is a need for a system that analyzes students' mental health using existing data 1 and provides meaningful insights.

1.2 Objectives

The main objectives of this project are:

• To use an existing dataset—such as the Student Mental Health Dataset—obtained via Kaggle to examine the mental health of students. The dataset used in this

project is publicly available on Kaggle: https://www.kaggle.com/datasets/shariful07/student-mental-health

- To find trends in the relationship between mental health and academic pressure.
- To build a system that predicts mental health levels.
- To provide recommendations based on the results.
- To create a simple and user-friendly system.

1.3 Expected Outcomes and Deliverables

The system is expected to provide:

- Mental health classification (Normal, Mild, Moderate, High Risk)
- Clear explanation of results
- Recommendations for improving mental health
- Data-driven insights

1.4 Motivation and Relevance

This project is motivated by the need to provide a data-driven solution that analyzes student mental health using publicly available datasets such as the Student Mental Health Dataset from Kaggle. By identifying pat- terns between academic workload and psychological indicators, the system helps students become more aware of their condition. The project also aligns with the UAE Vision 2031, which emphasizes improving quality of life, well-being, and the use of technology to support society. By using data analysis and machine learning, this system contributes to promoting mental health awareness and supporting students in managing stress effectively. Overall, the project aims to provide a simple, practical, and accessible solution that can positively impact students and improve their well-being.

1.5 Report Structure Summary

This report is organized into several chapters that explain the project in detail.

- Chapter 1 introduces the project, including the problem statement, objectives, expected outcomes, motivation, and overall structure of the report.

- Chapter 2 presents the background and related work, including existing systems and a gap analysis to highlight the need for the proposed solution.
- Chapter 3 describes the system requirements and analysis, including functional and non-functional requirements, tools and technologies, and feasibility considerations.
- Chapter 4 presents the conceptual design of the system, including system architecture, data design, and user interface planning.
- Chapter 5 discusses the professional, ethical, and sustainability considerations related to the project.
- Chapter 6 outlines the project management plan, including team structure, work breakdown structure, and project timeline.
- Chapter 7 presents the preliminary implementation of the system, including the prototype and initial results (if available).
- Chapter 8 provides the conclusion and summarizes the work completed in SGP1, along with future work planned for SGP2.
- Chapter 9 presents the statement of contribution, detailing the roles and contributions of each team member.

Chapter 2

Background and Related Work

2.1 Literature Review

Several existing mental health platforms have been developed to support users through applications such as Headspace which provides features such as meditation, stress relief management, and sleep support, it works becuase it has a simple and clean UI, and it focuses on daily habits (Headspace, 2012). Calm is another app that provides sleep stories, breathing exercises, and relaxation tools. It works because it targets stress and sleep, not just mentsl illness (Calm, 2017). Woebot provides a Chatbot to talk to an artificial intelligent "virtual friend", tracking moods, and giving CBT (Cognitive Behavior, Therapy) based responses. It works because because it feels like talking to someone and is always available. MindDoc provides daily questions, mood tracking, reports over time. It works because it has structured assessments and shows peogress. These tools encourage regular interaction and help users become more aware of their mental health.

In the UAE, platforms such as Pura, How is life and other digital mental health services emphasize accessibility, privacy, and remote support, allowing users to seek help more conveniently by offering online access to psychologist and peer to peer support (Gulf News, 2024). Pura offers online access to psycologists, and users can get help remotely anytime. It works because it is accessible, and real professionals are available (Pura, 2023). How is life is a service for any student that provides peer to peer support and allows students to talk to other students. It works because people feel more comfortable talking to others like them (How is life, 2024). This is especially important in regions where there is stigma around mental health which may prevent individuals from seeking in person support.

Overall, these systems demonstrate the growing importance of digital mental health solutions and highlighting key features such as accessibility, self monitoring, and personalized support.

2.2 Gap Analysis

Although existing applications offer useful features, many of them are designed for general users rather than specifically targeting university students. These systems often focus on general wellbeing and may not address unique challenges faced by students, such as academic stress, workload, and exam pressure.

Current solutions may not fully consider academic related factors such as credit hours, study load, exam schedules, and time management, which are major contributors to stress among university students. And most platforms don't integrate academic data with mental health assessments. As a result the analysis provided by these systems may not accurately reflect the students' stress patterns, limiting their ability to provide personalized and adaptive recommendations.

Our system aims to address these limitations by focusing specifically on university students. It combines academic data, such as schedules and exams, with questionnaire responses to provide a more accurate and personalized assessment of mental health risk levels.

Features/Systems	Headspace	Calm	Woebot	MindDoc	Pura	How is life	Our system
Target users	General users	General users	General users	General users	General users	Students	University students
Questionnaire	✗	✗	✓	✓	✓	✗	✓
Mood tracking	✗	✗	✓	✓	✓	✗	✓
AI analysis	✗	✗	✓	✓	✓	✗	✓
Professional support	✗	✗	✗	✗	✓	✗	✗
Peer support	✗	✗	✗	✗	✗	✓	✗
Personalized recommendations	✓	✓	✓	✓	✓	✗	✓
Integration of academic data	✗	✗	✗	✗	✗	✗	✓
Focus on academic stress	✗	✗	✗	✗	✗	✓	✓

Figure 2.1: Comparison Table of Similar Systems

2.3 Target Users or Stakeholders

The target users of the system are university students, specifically those experiencing academic stress, workload pressure, and challenges related to time managements and exams. The system is going to be designed to support students in understanding and managing their mental health in relation to their academic responsibilities.

Secondary stakeholders include universities, student counseling centers, and academic advisors, and educators. These stakeholders may benefit from insights into common student wellbeing concerns and stress patterns, which can help improve student support services and academic planning.

Chapter 3

Requirements and System Analysis

3.1 Functional Requirements

The following functional requirements describe what the system will do:

- FR 1 (User Management):
  - FR 1.1: The system needs to let people sign up for an account, so it should have a way for users to register.
  - FR 1.2: The system shall allow users to log in securely. The system should let users log out when they're done.
- FR 2 (Academic Data Input):
  - FR 2.1: The system shall allow users to enter their registered credit hours.
  - FR 2.2: The system shall allow users to input their weekly university schedule.
  - FR 2.3: The system shall allow users to enter their daily study hours.
  - FR 2.4: The system shall allow users to update their academic data.
- FR 3 (Exams Information):
  - FR 3.1: The system shall allow users to enter their final exam schedule.

  - FR 3.2: The system shall allow users to input the number of exams per week. The system needs to keep track of all the information related to exams.

- FR 4 (Questionnaire):
  - FR 4.1: The system shall display a mental health questionnaire related to academic stress.
  - FR 4.2: The system shall require all questions to be answered before submission. The system needs to be able to save the answers people give to questionnaires.

- FR 5 (Processing and Prediction):
  - FR 5.1: The system shall calculate a mental health score based on academic data and responses.
  - FR 5.2:The system shall use a machine learning model (scikit-learn) to classify the result.

- FR 6 (Results and Recommendations):
  - FR 6.1: The system shall display the mental health result.
  - FR 6.2: The system shall provide recommendations based on academic workload and exams.

- FR 7 (History and Tracking):
  - FR 7.1: The system needs to save the results in a database, and it will use MySQL to do this.
  - FR 7.2: The system needs to be able to save the readings that users have done, and also get them back when needed.
  - FR 7.3:The system shall allow users to view previous results.

- FR 8 (Smart Analysis):
  - FR 8.1:The system shall analyze the student's credit hours and schedule.
  - FR 8.2: The system shall analyze the final exam schedule.
  - FR 8.3: The system shall detect schedule overload.
  - FR 8.4: The system shall identify high stress periods (e.g., exam weeks).

3.2 Non-Functional Requirements

The following non-functional requirements define the system quality and performance:

- NFR 1 (Security):
  - NFR 1.1: The system shall store data securely in the database (MySQL).
  - NFR 1.2: The system shall ensure data confidentiality by restricting access to authorized users only.
- NFR 2 (Usability):
  - NFR 2.1:The system shall be simple and easy to use.
  - NFR 2.2: The system shall provide clear output results.

3.3 Applicable Standards and Constraints

The following standards and constraints apply to the system:

- AS 1 (Standards):
  - AS 1.1: The system follows NIST standards to ensure cybersecurity and protection of user data.
  - AS 1.2: The system follows ISO/IEC 27001 standards for information security management.
  - AS 1.3: The system follows IEEE standards for software development and system design.
- AS 2 (Ethical requirement):
  - AS2.1: The system shall protect user privacy and ensure data confidentiality.
  - AS 2.2: The system must clearly indicate that the results are not a medical diagnosis.
- AS 3 (Constraints):
  - AS 3.1: The system depends on available dataset (survey-based data).

  - AS 3.2: The project has a limited timeline, which is restricted to the academic semester.
  - AS 3.3: This system is not used for medical purposes or in clinical settings.

3.4 Tools and Technologies

The following tools and technologies are used in this project:

- TT 1 (Programming Language):
  - TT 1.1: Python is used for system development and implementation.
  - TT 1.2:We picked Python because it's easy to use, works well, and is really good at getting things done. ports machine learning.
- TT 2 (Web Framework):
  - TT 2.1: A web framework such as Django is used to develop the web-based application.
- TT 3 (Database):
  - TT 3.1: We use MySQL to keep all the user information and system results in one place.
  - TT 3.2: We picked MySQL because it's a reliable and secure option, and lots of people use it. used for structured data.
- TT 4 (Machine Learning):
  - TT 4.1: Scikit-learn is used to build the prediction model.
  - TT 4.2: It is chosen because it is easy to use and suitable for classification tasks.
- TT 5 (Development Tools):
  - TT 5.1: We use Visual Studio Code for all our coding and development needs.
  - TT 5.2: GitHub is used for version control and collaboration.
  - TT 5.3: We picked these tools because they're easy to use and support teamwork.

3.5 Feasibility and Risk Analysis

Feasibility and Risk Analysis The next part looks at whether the system can actually work and what might happen if it's used. risks:

- F 1 (Feasibility):
  - F 1.1 (The system is feasible using Python, MySQL, and scikit-learn, as these tools are simple and suitable for development.
  - F 1.2 (Economic Feasibility): The system is cost-effective since all tools used are free and open-source.
  - F 1.3 (Schedule Feasibility): The system can be completed within the academic semester timeline.
- F 2 (Risks):
  - F 2.1 (Technical Risk): The machine learning model may have low accuracy due to limited dataset. Mitigation: Use sufficient and clean data to improve model performance.
  - F 2.2 ((Schedule Risk): Limited time may affect project completion. Mitigation: Follow a clear plan and divide tasks among team members.
  - F 2.3 (Data Privacy Risk): User data may be exposed if not handled properly. Mitigation: Store data securely and restrict access to authorized users only.
  - F 2.4 (User Misunderstanding Risk): Users may think the system gives medical-diagnosis. Mitigation: Clearly state that the system provides recommendations only.

Chapter 4

Conceptual Design

This section describes how your system will function, the technologies you plan to use, and how the design meets project objectives and professional standards.

4.1 System Design and Architecture

The suggested system is intended to examine students' mental health using academic data and questionnaire answers. The system's basic architecture consists of the following key elements:
-User Interface (Web Application)
-Backend Processing
-Machine Learning Model
-Database
The system works as follows:
1-The user logs into the system.
2-The user enters required data such as study schedule and answers the questionnaire.
3-The data is sent to the backend for processing.
4-The machine learning model analyzes the data and predicts the mental health level.
5-The result is stored in the database. The system displays the result and recommendations to the user.
6-A sequence diagram is used to illustrate the interaction between system components.

4.1.1 System Design and Architecture

This system is created to process users' data to make predictions about their mental well-being using a well-defined process. This system mainly consists of the user, the web application, the back end, the machine learning algorithm, and the database.
Sequence diagrams show the collaboration of all these parts by demonstrating their interactions.

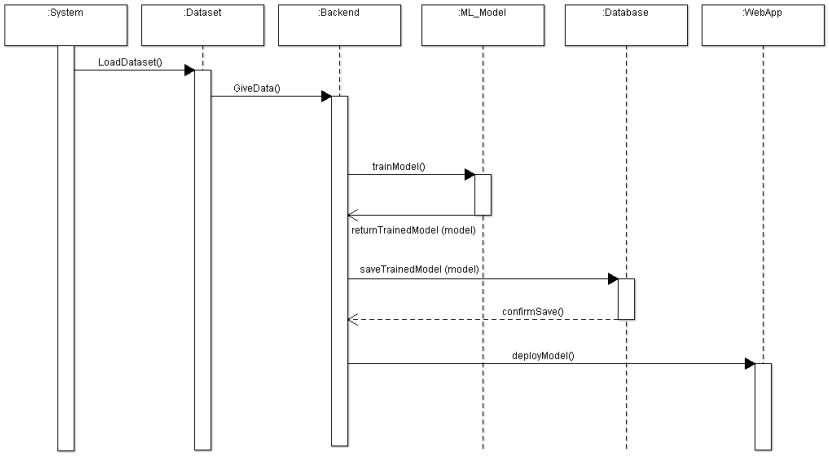


Figure 4.1: Sequence Diagram of the Model's Training and Deployment

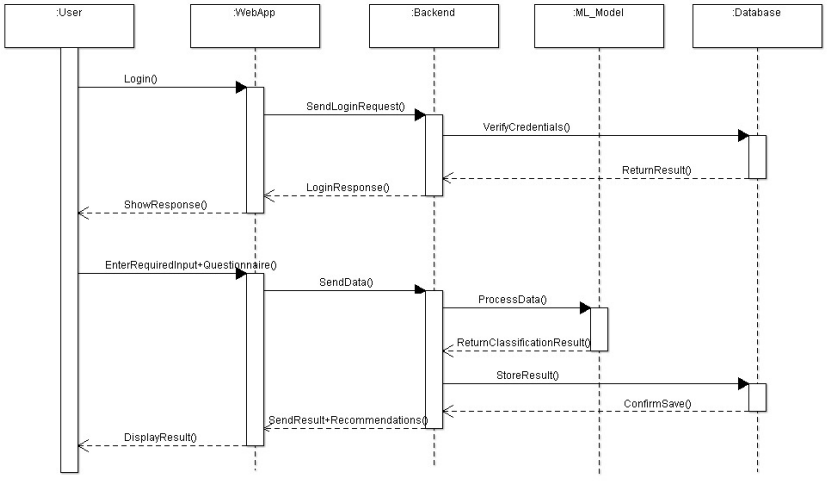


Figure 4.2: Sequence Diagram of the User System Interaction

The process begins with the user logging in. The login request is sent to the

backend, where it validates the login details with the database. If the login is successful, then the user fills in the relevant data on the form.
After that, the information reaches the backend, where it undergoes processing before being passed to the machine learning model. Classification occurs, and the results are stored in the database before being returned to the user along with recommendations.

4.2 Technologies and Implementation Approach

The system uses the following technologies:
-Programming Language: Python
-Machine Learning: scikit-learn
-Database: MySQL
-Frontend: Basic Web Interface
The reason behind choosing Python is its ability to process data efficiently and effectively. The prediction algorithm is designed using scikit-learn library. Data collected from users and questionnaires' output will be stored in the MySQL database.
These technologies are appropriate because to their simplicity, dependability, and widespread use in practical applications. Additionally, they facilitate system scalability and performance.

4.3 Data and Interface Design

The system data is organized into several main entities, including:
-User
-Study Schedule
-Questionnaire
-Responses
-Results
The entity relationship diagram explains the relationships between these entities, showing how user information, their reactions, and the results are stored and interconnected.
The interface of the system is designed in such a way that it is simple and easy to understand. It consists:
-Login page
-Questionnaire page
-Result page

The system focuses on usability, allowing users to easily enter data and view results without complexity.

4.3.1 Data and Interface Design

The data concerning the system includes some main objects such as the User, Study Schedule, Questionnaire, Responses, and Results. The data described is stored within the database and is related to each other in accordance with established relations.
The ER model illustrates how this data is structured and relates to each other.

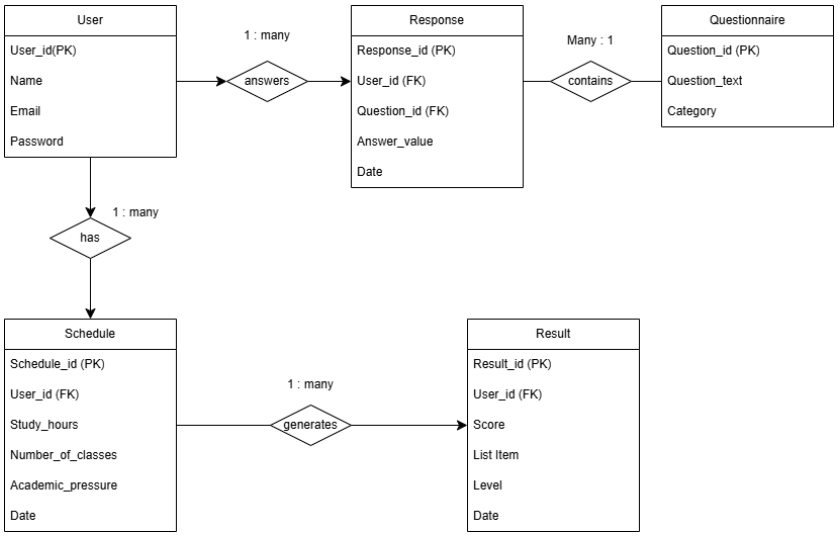


Figure 4.3: ER Diagram of the System

The design is intended to be user-friendly and uncomplicated. It contains screens such as the login screen, the questionnaire screen, and the result screen. The system focuses on simplicity, which allows users to input data and view the results without facing any complications.

4.4 Ethical and Innovative Aspects

By safeguarding individual data and safely storing information, the system guarantees user privacy. Sensitive mental health data is handled with care, and the system only offers broad assessment and recommendations rather than a medical diagnosis.

From an ethical standpoint, the system seeks to assist pupils without endangering or stressing them. Instead of making important choices, it offers direction.
In order to provide a more accurate analysis, the initiative also incorporates novel elements like combining academic data with mental health indicators. The system is improved by using machine learning to predict mental health levels based on student data collected from Kaggle.
All things considered, the system is made to be responsible, secure, and helpful to users.

Chapter 5

Professional, Ethical, and Sustainability Considerations

5.1 Preliminary Cost Estimation and Justification

- The system is cos effective as it relies on software tools and existing resources:
  - Hardware: Personal computers for development (Already available).
  - Software: Python, Django, MySQL, and scikit (All are open source and free)
  - Resources: Visual Studio Code and GitHub (Both are free)
  - Time: Development, testing, and research efforts throughout the semester.

5.2 Relevant Codes of Ethics and Moral Frameworks

This project follows the professional ethical guidelines, such as the ones by IEEE and ACM, in the areas of data privacy, user safety, and responsible computing.

Key ethical considerations are:

- User privacy: All user data must be stored securely and accessed only by authorized users.
- Confidentiality: Sensitive mental health related data must not be shared or exposed.
- Transparency: The system clearly states it does not provide medical diagnosis but only general recommendations.
- Responsibility: The system avoids misleading users and ensures outputs are understandable and not harmful.

These ethical considerations align with the standards and constraints that are defined in Chapter 3 and by following them, the system promotes ethical and responsible use of technology.

5.3 Relevant Environmental Considerations

The system is environmentally friendly as it is a digital web based application, reducing the need for physical resources. Environmental considerations include: reduced paper usage, efficient use of digital resources, and no requirement for special hardware or energy consuming systems. Overall, the system has a low environmental impact.

5.4 Relevance to UAE and Region (Social, Cultural, and Political)

The project relies project aligns with the UAE's efforts to promote mental health awareness. According to official UAE government resources, mental health is recognized as an important part of overall wellbeing, and is undertaking new measures to address mental health issues and reduce the stigma associated with it (UAE Government, 2024). This reinforces the importance of developing accessible digital tools that support student mental health.

5.5 Other Issues and Constraints (where relevant)

Several constraints and considerations affect the system design. These factors were considered to ensure the system is safe, practical, and appropriate for its users:

- Data privacy constraint: The need to protect sensitive user data requires the implementation of secure storage and access control mechanics, which will influence the system's design.
- Legal consideration: The system must avoid being classified as a medical tool, which will limit the type of feedback and recommendation provided.
- Accuracy limitations: Results depend on user input and available data, which may affect reliability.
- Accessibility: The system should be easy to use for all students, regardless of technical skill.
- Time constraint: The project must be completed within the time of the semester.

Chapter 6

Project Management Plan

6.1 Team Structure

The project team consists of three members, all of us assigned specific roles and responsibilities based on our expertise to ensure effective collaboration and successful completion.

- Meera Ataya (Information Security) Responsible for system security, data privacy, and ethical compliance. Helps with system design, writing documentation, and making parts of the system that are needed for safe data handling.
- Fatima Al Souden (Computer Science) Responsible for system development and implementation. Helps create the key features of the system and makes it easier for system components to work together.
- Noura Semaihi (Computer Science) Responsible for data analysis and machine learning. Creates and trains the prediction model and helps to make sure it works with the system.

The team collaborates to ensure smooth progress, high quality results, and timely completion of the project.

6.2 Work Breakdown Structure (WBS)

Present main project tasks and their dependencies. State the methodology of testing your system:

1 Project Planning: In this initial phase, the project objectives, scope, and deliverables are defined. The team structure is established, and

Table 6.1: Team Organization

Task	Task Completed By
Chapter 1: Introduction	Fatima Al Souden
Chapter 2: Literature Review	Noura Semaihi
Chapter 3: Methodology	Meera Ataya
Chapter 4: Implementation	Fatima Al Souden
Chapter 5: Testing and Evaluation	Noura Semaihi
Chapter 6: Project Plan	Meera Ataya
Chapter 8: Conclusion	Meera Ataya
Chapter 9: Statement of Contribution	All Team Members

the development environment is prepared. This phase serves as the foundation for all subsequent activities.

2 Requirements Gathering: This phase focuses on identifying system requirements by collecting academic-related data such as credit hours, study hours, weekly schedules, and exam load. Functional and non-functional requirements are clearly defined based on user needs.

3 System Design and Architecture: In this phase, the system architecture is designed, including database structure (MySQL), system modules, and data flow. The user interface is also planned to ensure usability and efficient interaction.

4 Development and Implementation: This phase involves building the system using Python. The machine learning model is developed using scikit-learn, and all system components are integrated to process user input and generate stress level predictions.

5 Testing and Quality Assurance: To ensure the system's functionality, accuracy, and usability, testing procedures are applied. The system is tested using Black Box Testing, where inputs and outputs are validated without considering internal code structure. In addition, White Box Testing is applied during development to verify internal logic and ensure correct implementation.

6 Test Cases: Test cases are designed to evaluate system behavior under different scenarios. For example, valid academic data should produce correct stress level predictions, while missing or incorrect inputs should generate appropriate error messages. Different workload combinations are tested to verify classification accuracy.

7 Experiment Setup and Expected Behavior: The system is evaluated us- ing various datasets and questionnaire responses. The machine learning model processes the input data and classifies stress levels into categories such as normal, moderate, or high. The expected behavior is accurate classification and meaningful output based on user input.

8 tools to be used in testing: The testing process utilizes several tools, including Python for implementation, scikit-learn for model evaluation, MySQL for database testing, and Visual Studio Code for development and debugging.

6.3 Timeline / Gantt Chart

The Gantt chart (Figure 1) presents the project schedule for SGP1 during the Spring 2026 semester (14 January 2026 to 24 April 2026). It outlines the main phases, including project initiation, requirements gathering, system design, development, testing, and documentation. Each phase is scheduled to ensure efficient progress and timely completion within the academic time-frame.

6.3.1 SGP1 (Spring 2026)

This stage represents the current phase of the project and focuses on planning, analysis, and initial development. The major deliverables include:

- Project proposal and problem definition
- Requirements analysis and system specifications
- Literature review and background study
- System design and architecture
- Prototype development
- Final report and presentation

6.3.2 SGP2

for the coming semester (SGP2) will focus on completing the system and preparing to make the system ready.

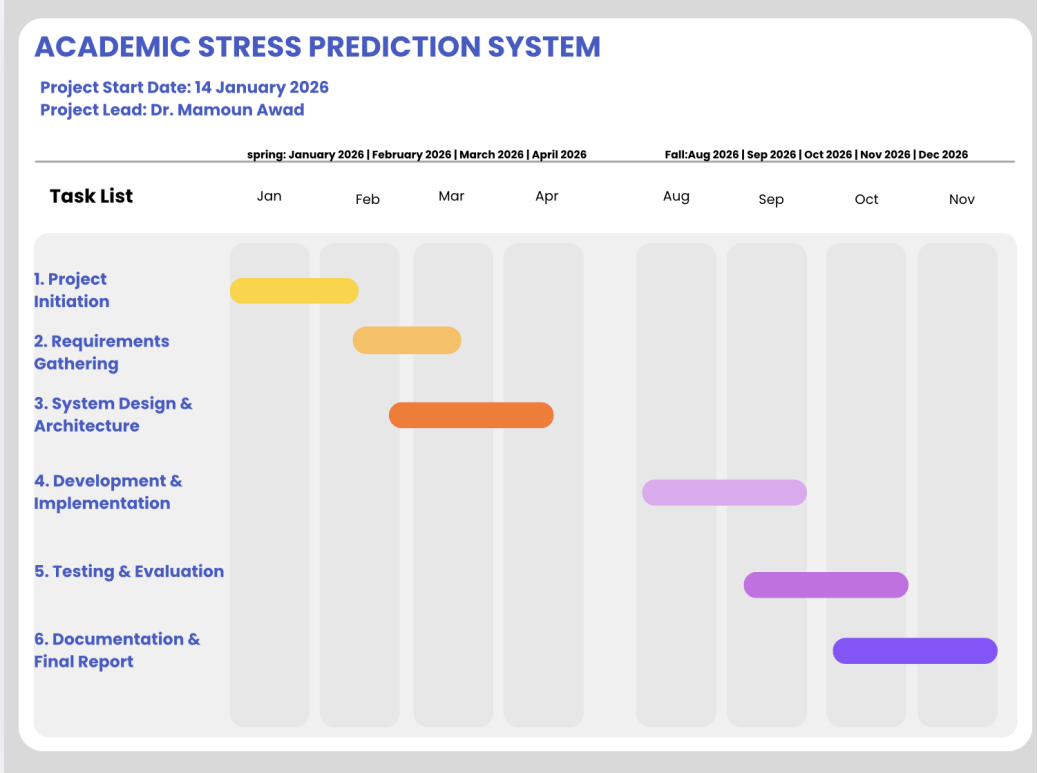


Figure 6.1: Gantt chart for the Academic Stress Prediction System project (UAEU Spring 2026).

• Development and Implementation (Weeks 1–4): Complete system development, including the implementation, integration of the machine learning model, and connecting the system components.
• Testing and Evaluation (Weeks 5–7): Conduct detailed system testing using black box and white box testing methodologies. Evaluate system accuracy, performance, and usability, and apply better improvements.
• Documentation and Final Report (Weeks 8–10): Prepare the final project report, including implementation details, results, and analysis. Complete the final presentation and system demonstration.

6.4 Project Milestones

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Chapter 7

Preliminary Implementation (Optional)

This part shows an early version of the Academic Stress Prediction System, which was made to show how the proposed solution would work and what its basic features would be. The SGP2 phase will see the full implementation, but an initial prototype has been made to show how the system will work in the real world. The prototype is a web-based interface that lets users easily interact with the system through a browser. It has a simple, easy-to-use design that lets students enter their academic information, such as credit hours, study hours, and exam information. The system also has a questionnaire module that lets you get more information about the student's stress level, study habits, and workload. These inputs are used to make a model of how the system processes data and gets it ready for prediction. The system makes a guess about the stress level and gives some basic advice after you enter the required information. This shows how the system is supposed to work: it looks at academic and questionnaire data to give the user useful information. The prototype is mostly about the following main parts:

- User input interface for academic data
- Stress assessment questionnaire
- Results dashboard displaying predicted stress level and recommendations

7.1 User Input Interface

The following figure presents the user input interface of the proposed system.

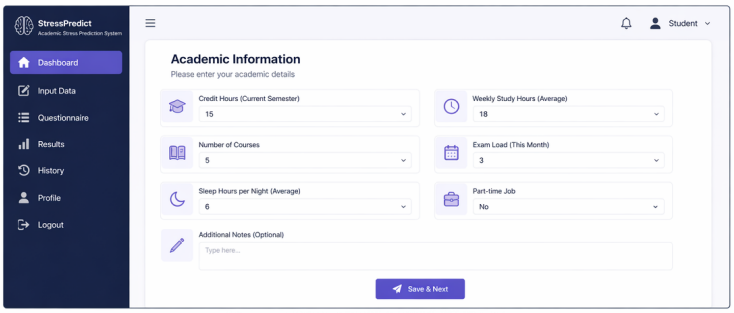


Figure 7.1: User Input Interface

Users can enter academic information like credit hours, study hours, and exam load through the interface. The system uses this data to figure out how stressed the student is and what will happen next.

7.2 Stress assessment questionnaire

The next figure shows how the system's questionnaire interface looks.
As shown in Figure 7.2, The students who use the service must answer a series of questions about their stress levels, study habits, and schoolwork. These answers are used to add more information to the prediction model.

7.3 Results dashboard

The following figure shows the results dashboard of the system.
As shown in Figure 7.3, The system shows the predicted stress level, a score indicator, and suggestions based on the user's input data.

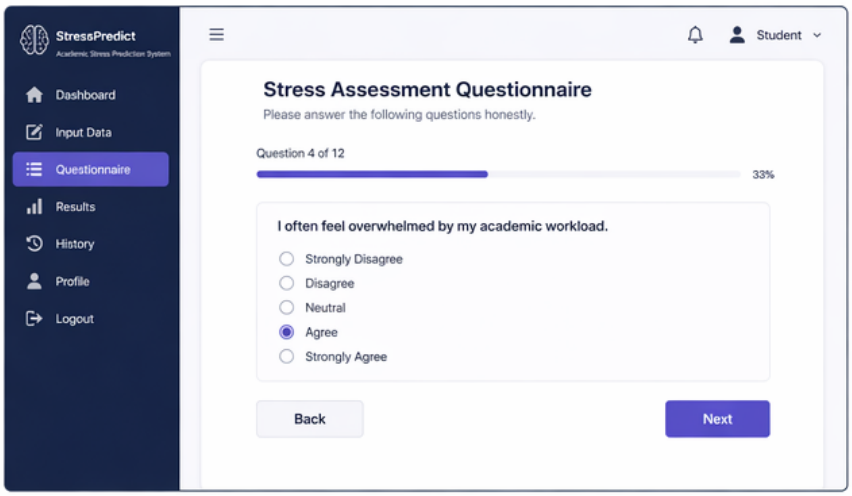


Figure 7.2: questionnaire

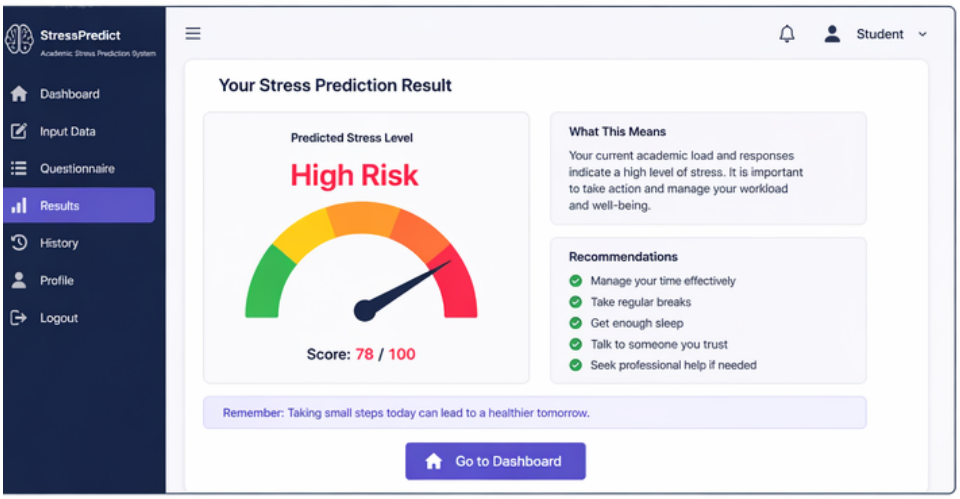


Figure 7.3: Results dashboard

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The goal of this project was to design and build an Academic Stress Prediction System that would use data to look at students' academic workload and predict their stress levels. Several important parts of the project were finished during the SGP1 phase. The issue was clearly defined, emphasizing the effect of academic stress on students' mental health and performance. A thorough review of the literature was done to find out what solutions are already out there and what needs to be improved. One of the biggest gaps was the lack of systems that take into account academic factors like credit hours, study time, and exam schedules. Also, the system requirements were spelled out, including both functional and non-functional ones. The overall design of the system, including the structure of the database and the planned user interface, was done. The right method was chosen, and the tools and technologies needed for development, like Python, MySQL, and scikit-learn, were found. A project plan was also made, which included the work breakdown structure and timeline, to make sure that progress was made in a planned and timely manner. They also made a simple prototype to show how the system will get user input and make stress level predictions. In general, the SGP1 phase laid a strong foundation for the project by clearly defining the system's design, structure, and approach.

8.2 Future Work (SGP2)

During the SGP2 phase, the focus will change to putting the whole system into action and testing it. we have chose some technologies that will be used to build and connect the system, and the machine learning model will be trained

on the right datasets to make predictions more accurate. The coming steps will be to build the user interface, connect it to the backend system, and make sure that data is processed and stored without any problems. Also, thorough testing will be done to check how well the system works, how accurate it is, and how easy it is to use. This means using testing methods like black box testing and checking the results that the model gives. Improvements and optimizations will be made to make the system more reliable and useful based on the test results. Finally, the system will be ready to be used and shown off. The final report and presentation will be done, and they will show the project's overall system, results, and contributions. The goal of SGP2 is to create the system accurate and easy to use that helps students better understand and deal with the stress of university.

Chapter 9

Statement of Contribution

This section presents the distribution of tasks accomplished during SGP1 along with the contribution percentage of each team member. The contributions are divided based on the tasks completed throughout the semester, its showing in 9.2 and Table 9.3 below.

Table 9.1: Student contribution distribution during SGP1

Task	Student ID	Contribution Percentage
T1	202100592	60%
	202113168	20%
	202213411	20%
T2	202100592	20%
	202113168	20%
	202213411	60%
T3	202100592	15%
	202113168	70%
	202213411	15%
T4	202100592	65%
	202113168	20%
	202213411	15%
T5	202100592	15%
	202113168	20%
	202213411	65%
T6	202100592	15%
	202113168	70%
	202213411	15%
T7	202100592	10%
	202113168	80%
	202213411	10%
T8	202100592	5%
	202113168	90%
	202213411	5%

Table 9.2: Student Overall contribution distribution during SGP1

Student ID	Contribution Percentage
202100592	30%
202113168	40%
202213411	30%

Table 9.3: Student Signatures

Student ID	Student Name	Signature
202100592	Fatima Al Souden	
202113168	Meera Ataya	
202213411	Noura Semaihi	

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